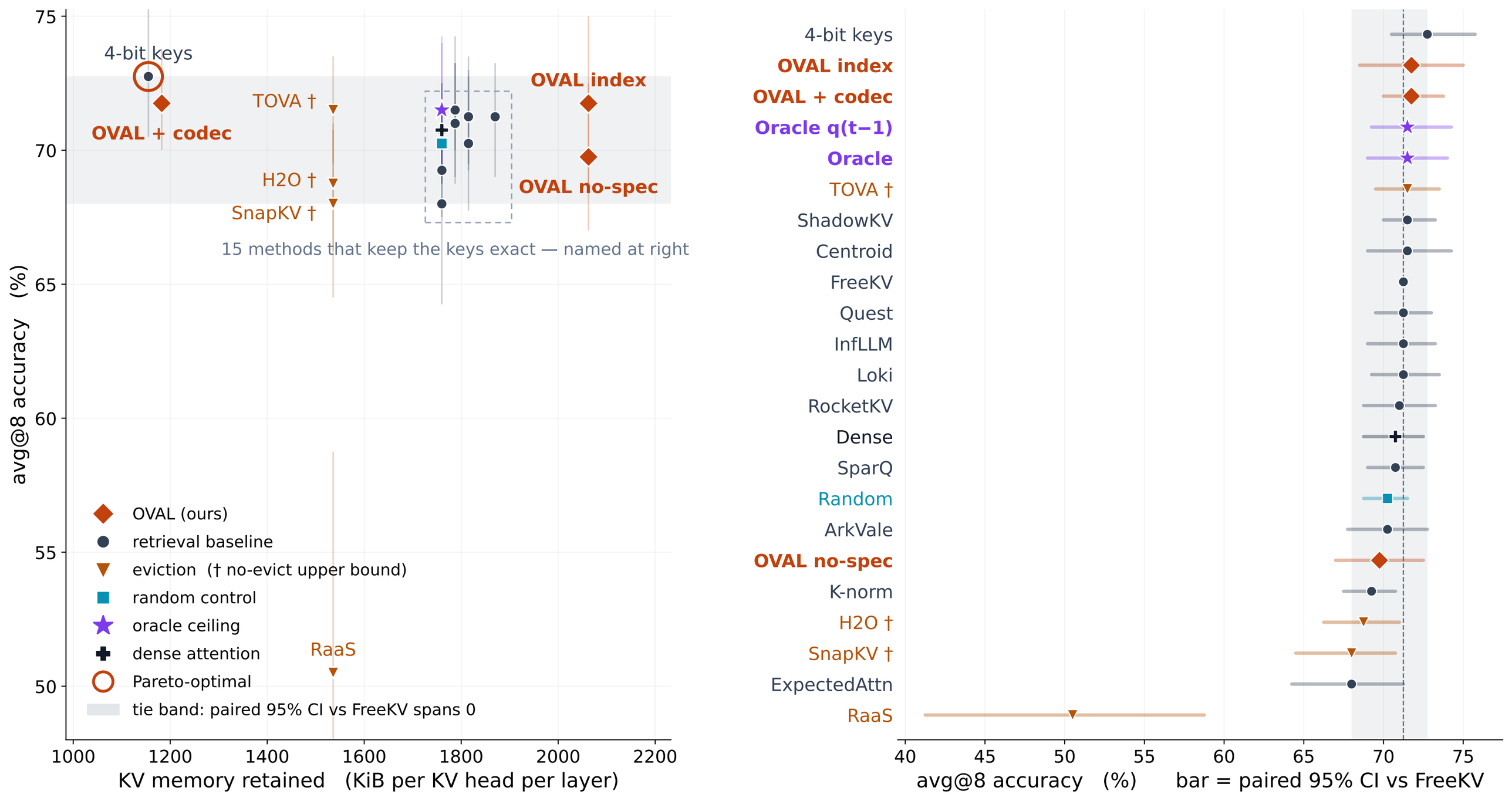
4-bit keys
OVAL + codec
TOVA †
H2O †
SnapKV †
OVAL index
OVAL no-spec
15 methods that keep the keys exact — named at right
RaaS
OVAL (ours)
retrieval baseline
eviction († no-evict upper bound)
random control
oracle ceiling
dense attention
Pareto-optimal
tie band: paired 95% CI vs FreeKV spans 0
avg@8 accuracy (%)
KV memory retained (KiB per KV head per layer)
4-bit keys
OVAL index
OVAL + codec
Oracle q(t−1)
Oracle
TOVA †
ShadowKV
Centroid
FreeKV
Quest
InfLLM
Loki
RocketKV
Dense
SparQ
Random
ArkVale
OVAL no-spec
K-norm
H2O †
SnapKV †
ExpectedAttn
RaaS
avg@8 accuracy (%)
bar = paired 95% CI vs FreeKV